HH-G-76C
January 5, 1970
SUPERSEDING
Fed. Spec. HH-G-76B
May 31, 1955

FEDERAL SPECIFICATION

GASKETS, ASBESTOS METALLIC CLOTH

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers metallic cloth gaskets for use with flanged joints.

1.2 Classification.

1.2.1 Types. Asbestos metallic cloth gaskets covered by this specification shall be of the following types, as specified (see 6.2).

Type I - Seamless.
Type II - Jointed.

1.2.2 Sizes. Gaskets covered by this specification shall be of the following sizes, as specified (see 6.2).

Size 1 - 3 by 4 inches I.D.
2 - 3 1/2 by 4 1/2 inches I.D.
3 - 11 by 15 inches I.D.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-B-665 - Boxes; Paperboard, Metal Stayed (including Stay Material).
PPP-B-676 - Boxes, Setup.
PPP-B-1055 - Barrier Material, Waterproofed, Flexible.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 191b - Textile Test Methods.
Fed. Test Method Std. No. 601 - Rubber; Method of Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifcations, Standards, and Handbooks and the Index of Federal Specifications and Standards from the established distribution points in their agencies.)

FSC 5330

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Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Classification Board:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Attention: Tariff Order Section, 1616 P Street, N.W., Washington, D.C. 20036.)

Uniform Classification Committee:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 Material. Gaskets shall be made from woven asbestos, wire-insertion cloth, treated with a friction compound composed of either natural (NR) or synthetic (SBR) rubber or a mixture of the two.

3.2 Construction.

3.2.1 General. Gaskets 1/8 inch and greater in thickness shall be folded. Gaskets 1/8 inch in thickness shall contain not less than 2 plies, and gaskets 3/16 inch and greater in thickness shall contain not less than 3 plies of asbestos wire-inserted cloth. The plies of cloth shall be uniformly impregnated with the friction compound.

3.2.2 Type I. Type I gaskets shall be seamless, composed of an integral piece of fabric in which the filler threads run continuously around the gasket; the warp threads shall run transversely. The fabric shall be folded on itself to form the desired thickness.

3.2.3 Type II. Type II gaskets shall be jointed. The cloth shall be cut on the bias and folded in the required shape and thickness after the ends have been securely joined together.

3.2.4 Dimensions. The dimensions and tolerances of the gaskets shall conform to those set forth in table I.

TABLE I. Dimensions and tolerances

Inside Dimensions		Flange Width	Thickness	Tolerances 1/	
Minor Axis Inches	X Major Axis Inches	Inches	Inches	Inches	Perimeter
11	15	1 1/4	1/4	1/32	1/32 on I.P. 2/ 20 inches or less.
3	4	5/8	1/4	1/32	
3 1/2	4 1/2	5/8	1/4	1/32	1/16 on I.P. more than 20 inches.

1/ No minus tolerances permissible.
2/ I.P. - Inside perimeter.

3.2.5 Gasket tape. When specified, gasket tape shall conform to the requirements for type II gaskets.

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3.2.6 Cloth. Each strand of the warp and filler in the cloth used in the construction of the gaskets shall consist of 2 plies of asbestos yarn and 1 copper zinc alloy wire twisted together.

3.2.6.1 Strands per inch. The asbestos wire-inserted cloth shall contain not less than 18 strands per inch in the warp and not less than 9 strands per inch in the filler.

3.2.6.2 Yarn. The yarn shall contain not less than 90 percent, by weight, long fiber asbestos. The yarn shall be not less than 12 percent, by weight, chemically combined water.

3.2.6.3 Wire. The copper zinc alloy wire used in the construction of the strands shall be 0.007 inch in diameter with a minus tolerance of 0.001 inch and a plus tolerance of 0.002 inch.

3.2.6.4 Weight. The weight of the asbestos wire-inserted cloth before application of the friction compound shall be not less than 2.25 pounds per square yard for 1/8-inch gaskets and 2.75 pounds per square yard for gaskets 3/16-inch and greater in thickness.

3.3 Friction compound. The friction compound shall be not more than 50 percent, by weight, of the finished gasket. The compound shall contain not more than 30 percent, by weight, of rubber hydrocarbon.

3.4 Effect of steam. Gaskets shall show no signs of disintegration of the friction compound, flowing, cracking or material hardening, when tested in accordance with 4.4.3.

3.5 Bending Strength. Gaskets shall not crack or show any signs of injury when tested as described in 4.4.4.

3.6 Rigidity. Gaskets shall be sufficiently rigid to retain their shape and size under ordinary conditions of handling both before and during installation.

3.7 Identification marking. Each bundle of gaskets shall be clearly and legibly labeled or marked with the manufacturer's name or trade-mark.

3.8 Workmanship. Gaskets shall be free from cracks, hardening, flowing in the friction compound or from any defects that will affect their function or serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling and inspection.

4.2.1 Lot. Unless otherwise specified (see 6.2), a lot shall consist of all sheets of the same type, size, and thickness offered for delivery at the same time.

4.2.2 Sampling for visual examination. Sampling of gaskets for the characteristics set forth in table II shall be in accordance with MIL-STD-105, using inspection level S-3. The sample unit shall be one sheet for sheets of gaskets.

4.2.3 Sampling for tests. Samples of gaskets shall be selected from each inspection lot for testing in accordance with MIL-STD-105, using inspection level S-2.

4.3 Examination.

4.3.1 Defects. Gaskets shall be examined for defects listed in table II. The acceptance quality level (AQL's) shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

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TABLE II. Classification of defects

Defects	Major	Minor
Length, not as specified.	X	
Type, size, and thickness not as specified.	X	
Cracks, hardening or flowing in the friction compound.	X	
Marking illegible or missing.		X
Damage or defects affecting function or serviceability.	X	
Damage or defects not affecting function or serviceability.		X

4.4 Test methods.

4.4.1 Strands.

4.4.1.1 Number. Determine the number of strands per inch in each of the warp and filling directions of the cloth in accordance with method 5050 of Fed. Test Method Std. No. 191.

4.4.1.2 Diameter. Using a micrometer or equivalent measuring device, measure the diameter of at least 5 wires to within 0.001 inch (see 3.2.6.3).

4.4.2 Weight of unfrictioned cloth. Determine the weight of unfrictioned cloth in accordance with method 5041 of Fed. Test Method Std. No. 191, except that the whole sample may be measured and weighed in place of smaller specimens cut from the sample (see 4.2.3).

4.4.3 Effect of steam. Determine the effect of steam on gaskets in accordance with method 7421 of Fed. Test Method Standard No. 601 with the following exceptions:

(1) The steam pressure shall be 275 ± 5 psi and the time of exposure to steam shall be 24 ± 1/4 hours.
(2) After exposure examine the specimen for disintegration of the friction compound, flowing, cracking and material hardening.

4.4.4 Bend test. Bend a sample 4 inches long flat on itself and examine it while in the bent position for cracking.

4.4.5 Chemical analysis.

4.4.5.1 Analyze for chemical composition in accordance with method 16601 of Federal Test Method Standard No. 601.

4.4.5.2 Determine the amount of rubber friction compound in the packing and the amount of rubber hydrocarbon in the friction compound in accordance with method 16611 of Federal Test Method Standard No. 601.

4.4.5.3 Determine the asbestos content of the yarns in the cloth and chemically combined water in asbestos in accordance with method 16651 of Federal Test Method Standard No. 601.

4.5 Examination of preparation for delivery. An exmination shall be made to determine that packaging, packing and marking as required by section 5 are complied with.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B or C as specified (see 6.2).

5.1.1 Level A and B.

5.1.1.1 Gaskets. Ten (10) gaskets of one stock number, shall be packaged flat in a box conforming to PPP-B-566, PPP-B-665, PPP-B-676, or PPP-B-636. Each box shall be secured to prevent accidental opening.

5.1.2 Level C. The gaskets shall be packaged in accordance with the supplier's commercial practice.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

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5.2.1 Level A. Unit packages, in quantities as specified (see 6.2) shall be packed in a box conforming to PPP-B-621, PPP-B-636 or PPP-B-640. The box shall be closed, waterproofed and strapped in accordance with appendix to the box specification.

5.2.2 Level B. Eight units packages of gaskets sized 11 by 15 inches (80 gaskets), or 24 unit packages of gaskets of other sizes (240 gaskets), or one stock number, shall be packed in a box conforming to PPP-B-636, class domestic. The box shall be closed in accordance with the appendix to the box specification.

5.3 Level C. The gaskets packaged as specified in 5.1 shall be packed in containers to assure carrier acceptance and safe arrival at destination in compliance with Uniform Freight Classification Rules or National Motor Freight Classification Rules.

5.4 Marking. In addition to any special marking specified in the contract or order the unit packages and shipping containers shall be marked in accordance with Fed. Std. No. 123 or MIL-STD-129 as applicable.

6. NOTES

6.1 Intended use. Gaskets covered by this specification are designed for use in boiler manholes, handholes and flanged joints and in applications using steam, gases, air and aqeous solutions other than strong mineral acids up to 300 psi and 600°F.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, symbol, and date of the specification.
(b) Types required (see 1.2.1).
(c) Size required (see table I).
(d) Size of lot, if different from 4.2.1.
(f) Selection of applicable level of packaging and packing required (see section 5).

MILITARY CUSTODIANS:

Army - MR
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Review activities:

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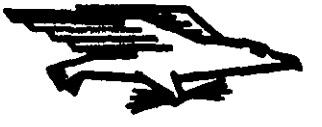
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SPECIFICATION COMMENT SHEET

BUDGET BUREAU NO. 29-R0175

INSTRUCTIONS

This form provides a way for users of this specification to inform the originator of problems encountered in its use. It is not to be used to request changes to accommodate proprietary features. All comments will be considered and appreciated, but please do not expect a reply. To comment: detach, complete, fold, staple, and mail.

NOTE: Comments on this form do not constitute or imply authorization to waive any part of the document or serve to amend contractual requirements.

1. SPECIFICATION

HH-G-76C - Gaskets, Asbestos Metallic Cloth

2. CONTRACT NO. *(If any)*	3. QUANTITY ON CONTRACT *(Optional)*	4. DOLLAR VALUE *(Optional)*

5. GENERAL NATURE OF PROBLEM *(e.g., inspection difficulties, manufacturers unable to meet tolerances, containers collapse under normal warehousing conditions, etc.)*

6. SPECIFIC REQUIREMENTS AFFECTED *(Include paragraph number and lines of wording)*

7. SPECIFIC PROBLEMS *(e.g. tests in 4.2.2 will not assure that the battery will last required time; temperature ranges in table 2 do not conform to commercially available items.)*

8. RECOMMENDATIONS

9. NAME OF MANUFACTURER, ASSOCIATION, GOVT., AGENCY, ETC.	10. ADDRESS *(Number, Street, City, State and Zip Code)*

11. NAME AND TITLE OF SUBMITTER	12. DATE

GSA DC 69-9761

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